

MAINT: Fix use-after-free bug in Py_FindObjects #15013

New issue

Merged

tupui merged 1 commit into `scipy:master` from `AnirudhDagar:fix_ndimage_refcount` on Nov 11, 2021

Conversation 3

Commits 1

Checks 0

Files changed 1

+0 -1

AnirudhDagar

commented on Nov 10, 2021

Member

Reference issue

Closes [gh-14713](#). See the issue for more details.

What does this implement/fix?

Remove the extra reference decrement which is not needed and may cause the ref count to become negative in case of a memory error (though that's not likely).

cc [@tupui](#) [@Snape3058](#)

Reviewers

tupui

Assignees

No one assigned

Labels

defect

scipy.ndimage

Projects

None yet

Milestone

1.8.0

Development

Successfully merging this pull request may close these issues.

BUG: Potential use-after-free bug in fu...

2 participants



MAINT: Fix use-after-free in `Py_FindObjects`

Verified

e9fad96

AnirudhDagar added the `scipy.ndimage` label on Nov 10, 2021

tupui

reviewed on Nov 10, 2021

View reviewed changes

tupui

left a comment

Member

My understanding of this was that `Py_XDECREF` would not do anything if there is NULL as opposed to `Py_DECREF` . But if we don't need this at all here (which seems to be the case) then why not remove this line.

AnirudhDagar

commented

Member

Author

on Nov 11, 2021

My understanding of this was that `Py_XDECREF` would not do anything if there is NULL as opposed to `Py_DECREF`

Yes that is true, but in this case, the result is not NULL if/when the `PyErr_Occurred` holds true. In that case, it is actually a dangling pointer since the ref count has been decremented to zero once and then again the second decrement still goes through (since it is not NULL) making the ref count -1. So we can just remove this extra DECREF which is not required anyway.

I believe this bug managed to get in at the first place in anticipation to counter the complimentary increment on L920:

[scipy/scipy/ndimage/src/nd_image.c](#)

Line 920 in [8ef5830](#)

```
920     Py_INCREF(result);
```

But in case of an error (Eg L891), the control is transferred directly to the exit label and that increment is never hit.



tupui merged commit `e32fc23` into `scipy:master`
on Nov 11, 2021



tupui commented on Nov 11, 2021 • edited

Member

Ok thanks [@AnirudhDagar](#) for the precisions. (Failures are not linked)



tupui added this to the **1.8.0** milestone on Nov 11, 2021



tupui added the **defect** label on Nov 11, 2021